

Name:

Date:

Period:

Chapters 4 and 5 Homework

Homework Check A (collected by Fall Break - Oct 10)

Reading

Please read the following on your own in the OpenStax textbook by the dates given. It will give good context for class discussion. Check off when you have completed them.

- ☐ 4.1 Development of the Force Concept Fri, Sept 26
- ☐ 4.2 Newton's First Law: Inertia Mon, Sept 29
- ☐ 4.3 Newton's Second Law: Concept of a System Tue, Sept 30
- ☐ 4.4 Newton's Third Law: Symmetry in Forces Tue, Sept 30
- ☐ 4.5 Normal, Tension, and Other Forces Fri, Oct 3

Problems and Conceptual Question

Get stamps from your instructor as you complete each of the following problems. The conceptual questions (CQ) require at least one sentence of explanation.

STAMPS WILL NOT BE GIVEN IF WORK IS NOT SHOWN ON A SEPARATE SHEET OF PAPER

Forces & Newton's Laws (5 POINTS)	Single Body (5 POINTS)
Multi Body (5 POINTS) HW Quiz on Tue, Oct 7.	

Equations

$$\Sigma \vec{F} = m\vec{a}$$

$$w = mg$$

Name:

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Chapters 4 and 5 Homework

Homework Check B (collected on Test Day - Tue, Nov 4)

Reading

Please read the following on your own in the OpenStax textbook by the dates given. It will give good context for class discussion. Check off when you have completed them.

☐ 5.1 FrictionTue, Oct 28

Problems and Conceptual Question

Get stamps from your instructor as you complete each of the following problems. The conceptual questions (CQ) require at least one sentence of explanation.

Friction (5 POINTS)	Forces at an Angle (5 POINTS)

Bonus Problems

P#3	P#9	P#21

Homework will be accepted for full credit until the test. Homework turned in after the test will be accepted for half credit until the next test. *Please remember that you will not be eligible to complete test corrections if you do not turn in your homework.*

Equations

$$\Sigma \vec{F} = m\vec{a}$$

$$w = mg$$

$$f_s \leq \mu_s N$$

$$f_k = \mu_k N$$

Questions and figures are from OpenStax *College Physics*, 2nd ed (Chapter 4) unless otherwise noted.

Forces & Newton's Laws

- (OpenStax P1) A 63.0-kg sprinter starts a race with an acceleration of 4.20 m/s^2 . What is the net external force on him?
- (OpenStax P10) A powerful motorcycle can produce an acceleration of 3.50 m/s^2 while traveling at 90.0 km/h . At that speed the forces resisting motion, including friction and air resistance, total 400 N . What is the magnitude of the force the motorcycle exerts backward on the ground to produce its acceleration if the mass of the motorcycle with rider is 245 kg ?
- Bonus!** (OpenStax P44) When starting a foot race, a 70.0-kg sprinter exerts an average force of 650 N backward on the ground for 0.800 s . (a) What is his final speed? (b) How far does he travel?

Conceptual Questions

- (OpenStax CQ2) What properties do forces have that allow us to classify them as vectors?
- (OpenStax CQ3) How are inertia and mass related?
- (OpenStax CQ5) Which statement is correct? (a) Net force causes motion. (b) Net force causes change in motion. Explain your answer and give an example.
- (OpenStax CQ8) Describe a situation in which the net external force on a system is not zero, yet its speed remains constant.
- (OpenStax CQ15) When you take off in a jet aircraft, there is a sensation of being pushed back into the seat. Explain why you move backward in the seat—is there really a force backward on you? (The same reasoning explains whiplash injuries, in which the head is apparently thrown backward.)
- (OpenStax CQ18) Why does an ordinary rifle recoil (kick backward) when fired? The barrel of a recoilless rifle is open at both

ends. Describe how Newton's third law applies when one is fired. Can you safely stand close behind one when it is fired?

- CQ7. (Original Question) A horse pulls a cart with a force of 400 Newtons forward. However, according to Newton's Third Law, whenever the horse pulls the cart, there is an equal and opposite reaction of the cart pulling the horse of 400 N backwards. Why do these two forces not cancel each other out giving to a net force of zero?

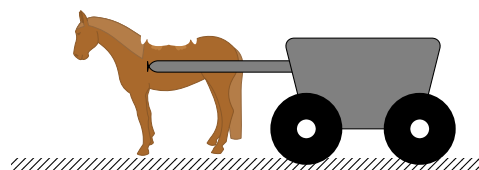


Figure 1: (Original Diagram) A horse and cart.

- CQ8. (Original Question) According to Newton's Third Law, when you jump off a desk, the force the Earth exerts on you, is equal to the force you exert on the Earth. However, it seems as if only you accelerate down. The earth doesn't seem to accelerate up. Why is this?
- CQ9. (Original question) If a large semi-truck gets into a head-on collision with a small car, which vehicle experiences the larger force? Which vehicle experiences the larger acceleration?

Single-Body

- (OpenStax P9) Suppose two children push horizontally, but in exactly opposite directions, on a third child in a wagon. The first child exerts a force of 75.0 N , the second a force of 90.0 N , friction is 12.0 N , and the mass of the third child plus wagon is 23.0 kg . (a) What is the system of interest if the acceleration of the child in the wagon is to be calculated? (b) Draw a free-body diagram, including all forces acting on the system. (c) Calculate the acceleration. (d) What would the acceleration be if friction were 15.0 N ?

⁴(c) 0.130 m/s^2 ; (d) 0 m/s^2

¹ 265 N
² 1260 N

5. (OpenStax P5) In Figure 2 below, the net external force on the 24-kg mower is stated to be 51 N. (a) If the force of friction opposing the motion is 24 N, what force F (in newtons) is the person exerting on the mower? (b) Suppose the mower is moving at 1.5 m/s when the force F is removed. How far will the mower go before stopping?



Figure 2: (OpenStax Fig 4.7) The net force on a lawn mower is 51 N to the right.

6. (OpenStax P13) The weight of an astronaut plus their space suit on the Moon is only 250 N. How much do they weigh on Earth? What is the mass on the Moon? On Earth? (The acceleration of gravity on the moon is 1.62 m/s^2)
7. (OpenStax P14) Suppose the mass of a fully loaded module in which astronauts take off from the Moon is 10,000 kg. The thrust of its engines is 30,000 N and the acceleration of gravity on the moon is 1.62 m/s^2 . (a) Calculate its the magnitude of acceleration in a vertical takeoff from the Moon. (b) Could it lift off from Earth? If not, why not? If it could, calculate the magnitude of its acceleration.
8. (Original Question) You are standing on a scale in an elevator. If your mass is 67 kg, what force would the scale read if the elevator accelerates up at a rate of 2.7 m/s^2 ? Remember, a scale measures normal force, not weight. Make sure to draw and label the free-body diagram.
9. **Bonus!** (OpenStax P23) A 5.00×10^5 -kg rocket is accelerating straight up. Its engines produce $1.250 \times 10^7 \text{ N}$ of thrust, and air resistance is $4.50 \times 10^6 \text{ N}$. What is the rocket's acceleration? Explicitly show how you follow the steps in the Problem-Solving Strategy for Newton's laws of motion.

⁵(a) 75 N; (b) 1.1 m

⁶1500 N

⁷(a) 1.38 m/s^2 ; (b) cannot take off

⁸837.5 N

Conceptual Question

- CQ10. (OpenStax CQ23) To simulate the apparent weightlessness of space orbit, astronauts are trained in the hold of a cargo aircraft that is accelerating downward at g . (a) Why will they appear to be weightless, as measured by standing on a bathroom scale, in this accelerated frame of reference? (b) Is there any difference between their apparent weightlessness in orbit and in the aircraft?

Multi-Body

10. (OpenStax P22) Consider the baby being weighed in Figure 3. (a) What is the mass of the child and basket if a scale reading of 55 N is observed? (b) What is the tension T_1 in the cord attaching the baby to the scale? (c) What is the tension T_2 in the cord attaching the scale to the ceiling, if the scale has a mass of 0.500 kg? The masses of the cords are negligible.

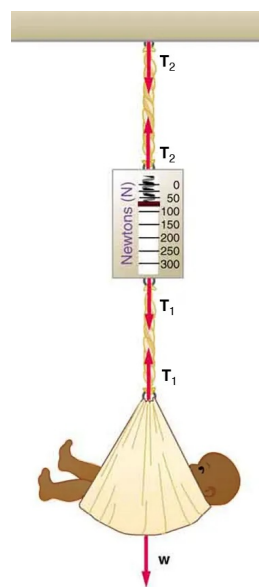


Figure 3: (OpenStax Fig 4-33) A baby is weighed using a spring scale.

¹⁰(a) 5.6 kg; (b) 55 N; (c) 60 N

11. (OpenStax P34) Figure 4 shows Superhero and Trusty Sidekick hanging motionless from a rope. Superhero's mass is 90.0 kg, while Trusty Sidekick's is 55.0 kg, and the mass of the rope is negligible. (a) Draw a free-body diagram of the situation showing all forces acting on Superhero, Trusty Sidekick, and the rope. (b) Find the tension in the rope above Superhero. (c) Find the tension in the rope between Superhero and Trusty Sidekick. Indicate on your free-body diagram the system of interest used to solve each part.
12. (Original question) The top rope in Figure 4 is attached to the bottom of an elevator that begins to accelerate upward. The tension of the top rope increases to 1600 N. (a) Calculate the tension on the rope between Superhero and Trusty Sidekick. (b) Calculate the acceleration of the two heroes.



Figure 4: (OpenStax Fig 4-38) For problems 11 & 12. Superhero (at the top) is 90.0 kg; Trusty Sidekick (on the bottom) is 55.0 kg. The ropes are massless.

13. (Original question) Two blocks of masses $m_1 = 5.0$ kg and $m_2 = 3.0$ kg are connected by a light, inextensible string that passes over a frictionless pulley. The system is released from rest.
- Draw a free-body diagram for each block.
 - Determine the acceleration of the system.
 - Find the tension in the string.

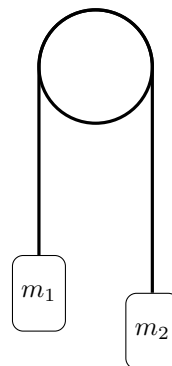


Figure 5: (Original Diagram) Figure for problem 13.

Conceptual Question

- CQ11. (Original Question) Consider the two boxes of masses $m_1 > m_2$ attached to each other by a massless cord shown in Figure 6. In both cases, the system is pulled by 75 N of force. In which orientation is the tension force larger? Explain. You may neglect friction.

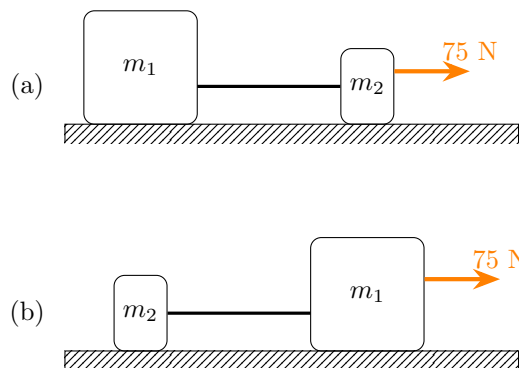


Figure 6: (Original Diagram) Figure for CQ11.

¹¹(b) 1420 N; (c) 539 N

¹²(a) 607 N; (b) 1.23 m/s²

¹³(b) 2.45 m/s²; (c) 36.75 N

Friction

Problems

System	μ_s	μ_k
Rubber on dry concrete	1.0	0.7
Rubber on wet concrete	0.7	0.5
Wood on wood	0.5	0.3
Waxed wood on wet snow	0.14	0.1
Metal on wood	0.5	0.3
Steel on steel (dry)	0.6	0.3
Steel on steel (oiled)	0.05	0.03
Teflon on steel	0.04	0.04
Shoes on wood	0.9	0.7
Shoes on ice	0.1	0.05
Ice on ice	0.1	0.03
Steel on ice	0.04	0.02

Table 1: (OpenStax Table 5.1) Coefficients of Static and Kinetic Friction

14. (OpenStax Ch 5, P1) A physics major is cooking breakfast when he notices that the frictional force between his steel spatula and his Teflon frying pan is only 0.200 N. Knowing the coefficient of kinetic friction between the two materials, he quickly calculates the normal force. What is it? (Use the coefficients of friction provided in Table 1)
15. (OpenStax Ch 5, P4) Suppose you have a 120-kg wooden crate resting on a wood floor. (a) What maximum force can you exert horizontally on the crate without moving it? (b) If you continue to exert this force once the crate starts to slip, what will the magnitude of its acceleration then be? (Use the coefficients of friction provided in Table 1)
16. (Original Question) A 7-kg wood box is pushed across a horizontal wood floor. (a) Once there is no longer any applied force, what is the box's acceleration? (b) How far does the box slide before coming to a complete stop if its initial velocity is 4.4 m/s?

¹⁴5.00 N

¹⁵5.00 N

¹⁵(a) 588 N; (b) 1.96 m/s²

¹⁶(a) -2.94 m/s²; (b) 3.29 m

17. (Original question) A metal block of mass $m_1 = 25$ kg rests on a horizontal wood table. The block is connected by a light, inextensible string that passes over a frictionless pulley to a hanging block of mass $m_2 = 10$ kg.

- (a) Draw a free-body diagram for each block.
 (b) Determine the acceleration of the system.
 (c) Find the tension in the string.

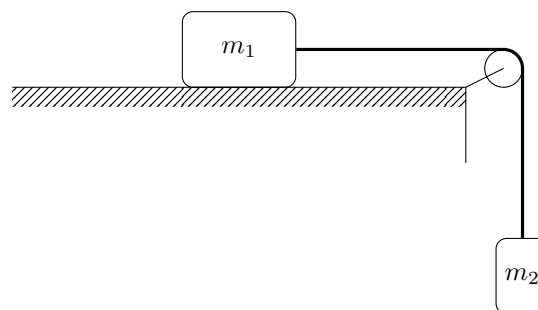


Figure 7: (Original Diagram) Figure for problem 17.

18. (OpenStax Ch 5, P5) (a) If half of the weight of a small 1,000-kg utility truck is supported by its two rubber drive wheels, what is the magnitude of the maximum acceleration it can achieve on dry concrete? (b) Will a 45-kg metal cabinet lying on the wooden bed of the truck slip if it accelerates at this rate?

¹⁷(b) 0.70 m/s²; (c) 91 N

¹⁸(a) 4.90 m/s²; (b) will not slip

Forces at an Angle

Problems

19. (OpenStax Ch 5, P21 with significant modification) A contestant in a winter sporting event pushes a 45.0-kg block of ice across a frozen lake as shown in Figure 8(a). A second contestant pulls it as in Figure 8(b). Both contestants exert a force $F = 15$ N. Calculate the acceleration of and normal force on each block.

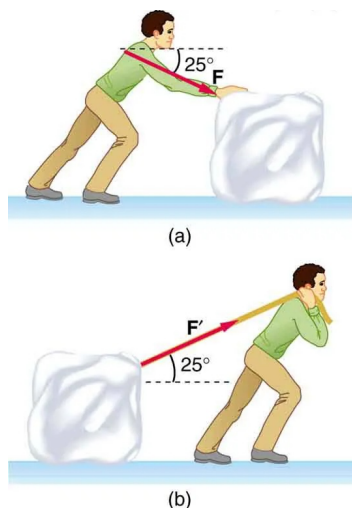


Figure 8: (OpenStax Fig 5.21) Which method of sliding a block of ice requires less force—(a) pushing or (b) pulling at the same angle above the horizontal?

20. (OpenStax Ch 5, P11) (a) Calculate the acceleration of a 70-kg skier heading down a 10° slope, assuming the coefficient of friction for waxed wood on wet snow from Table 1.
21. **Bonus!** (OpenStax Ch 5, P11) Find the angle of the slope down which the skier in the previous problem could coast at a constant velocity.
- CQ12. (Original Question) Look above at Figure 8. This time, assume that friction is *not* negligible. Which method of moving the block would be easier: pulling from in front or pushing from the back? Explain.

¹⁹(a) 0.302 m/s^2 , 434.7 N ; (b) 0.302 m/s^2 , 447.3 N

²⁰ 0.737 m/s^2

²¹ 5.71°

- CQ13. (Original Question) One block sits at rest on a level surface while a second identical block sits at rest on an incline. In which case is the normal force larger? Why?

Additional Practice Problems

These problems are not required and are not bonus. Worked out solutions are available on Schoology.

22. (Ch 4, P19 from Giancoli *Physics: Principles with Applications*, 7th ed.) The cable supporting a 2125-kg elevator has a maximum strength of 21,750 N. What maximum upward acceleration can it give without breaking?
23. (Ch 4, P37 from Giancoli) A force of 35.0 N is required to start a 6.0-kg box moving across a horizontal concrete floor. (a) What is the coefficient of static friction between the box and the floor? (b) If the 35.0-N force continues, the box accelerates at 0.60 m/s^2 . What is the coefficient of kinetic friction?
24. (Ch 4, P25 from Giancoli) One 3.2-kg paint bucket is hanging by a massless cord from another 3.2-kg paint bucket, also hanging by a massless cord, as shown in Figure 9. (a) If the massless buckets are at rest, what is the tension on each cord? (b) If the two buckets are pulled upward with an acceleration of 1.25 m/s^2 by the upper cord, calculate the tension on each cord.

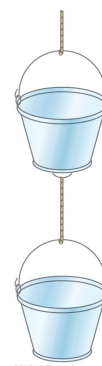


Figure 9: Fig 4-49 from Giancoli

²² 0.44 m/s^2

²³(a) $\mu_s = 0.60$ (b) $\mu_k = 0.53$

²⁴(a) 31.36 N and 62.72 N ; (b) 35.36 N and 70.72 N